QISKIT FALL FEST 2026 | KNUST EDITION

# A Decade in the Cloud: From Qubits to Global Access

The Evolution, Practical Breakthroughs, and Industry Scope of Quantum Technology

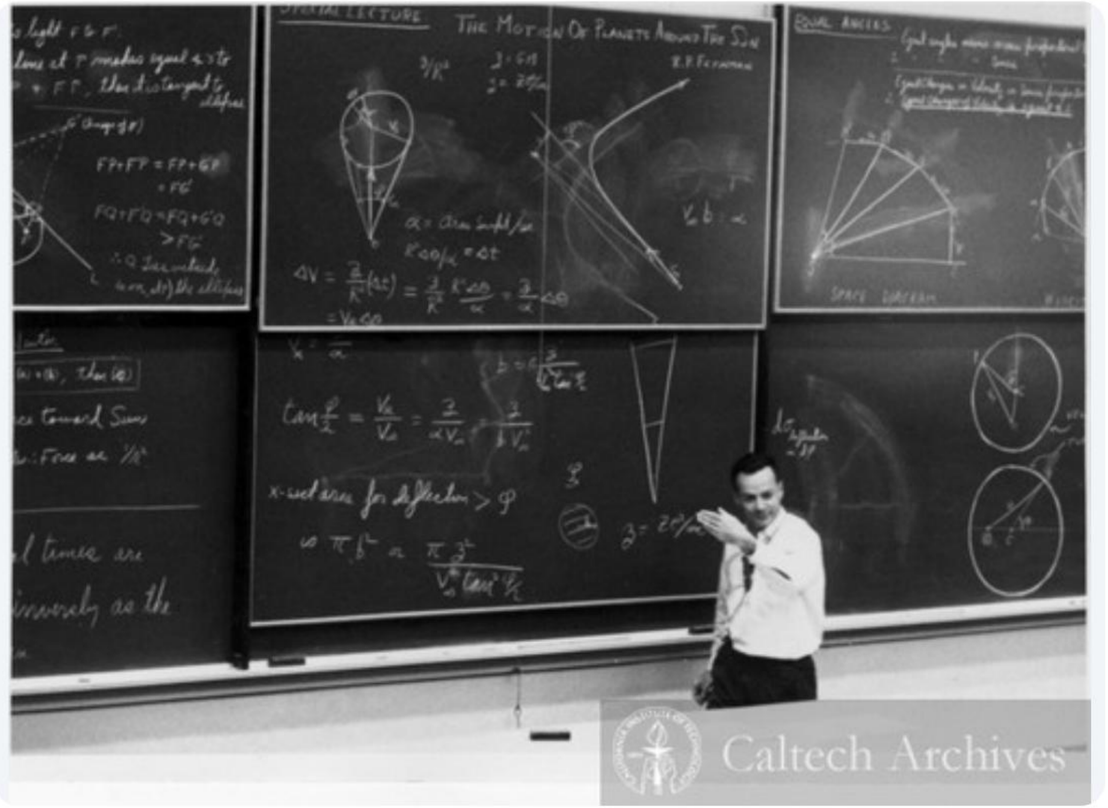
FOUNDATIONAL VISION

# The Spark: Simulating Nature

> "Nature isn't classical, dammit, and if you want to make a simulation of nature, you'd better make it quantum mechanical."

**Richard P. Feynman (MIT, 1981)**

Classical computers cannot scale to simulate quantum chemistry and materials efficiently.



HISTORICAL TRAJECTORY

# Core Milestones in Quantum Theory

## 1981

### Quantum Simulation

Feynman proposes physical quantum machines to model atomic systems.

## 1994

### Shor's Algorithm

Proves exponential speedup for integer factorization and cryptography.

## 1996

### Grover's Search

Establishes quadratic acceleration for unstructured database search.

ACCESSIBILITY REVOLUTION

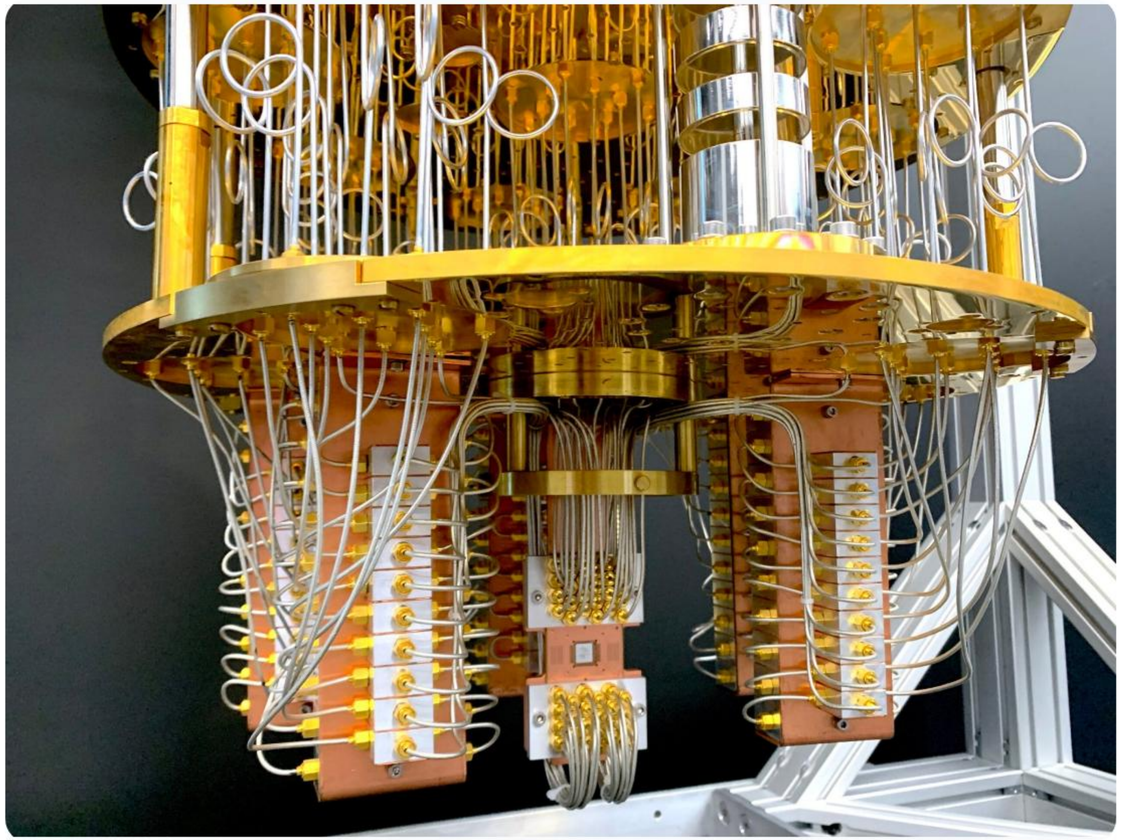
# May 2016: Quantum Hits the Cloud

Before 2016, quantum hardware was confined to specialized physics laboratories. Cloud connectivity opened real QPUs to every student and researcher.

**Global Access:** Running circuits on real quantum hardware via the internet.

**Open Source:** Launch of Qiskit for cross-platform circuit development.

**Community Growth:** Global student chapters and open scientific collaboration.



A DECADE OF PROGRESS

# Scaling Hardware: From Demos to Utility

**5**

**Qubits in 2016**

Proof of concept via IBM Quantum Experience

**127**

**Qubits in 2021**

IBM Eagle: Beyond brute-force exact simulation

**156+**

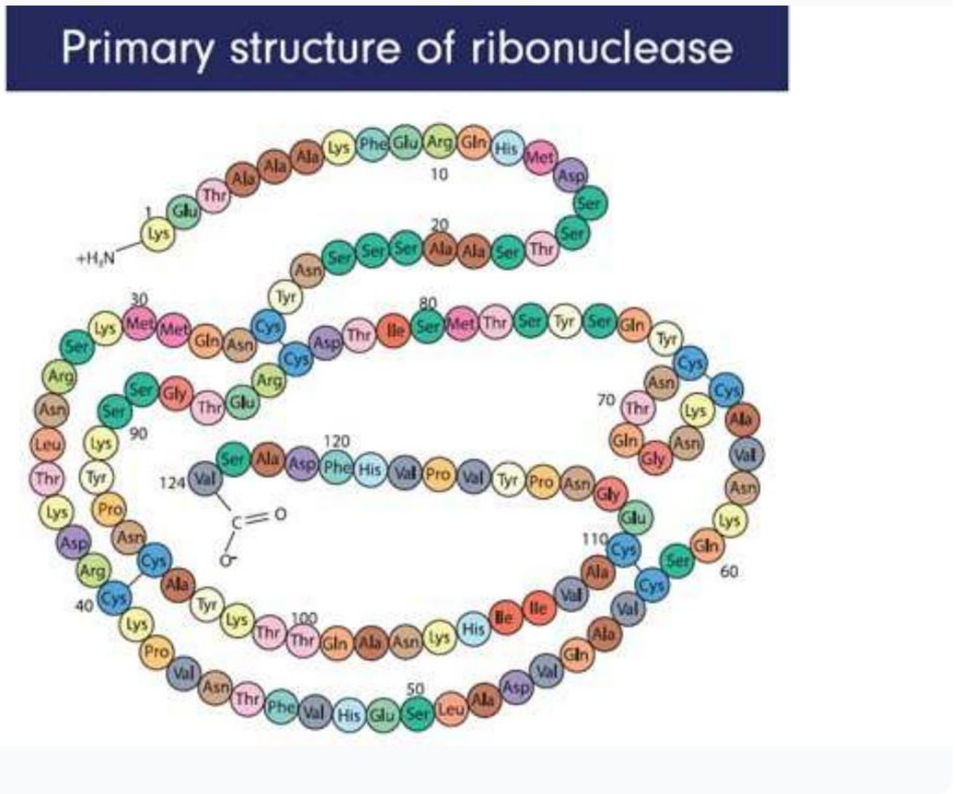
**Qubits Today**

IBM Heron: Low error rates for utility workloads

Shift from lab experiments to quantum utility across scientific domains.

APPLICATION DOMAIN

# Quantum Chemistry and Materials




Directly mapping electron orbital interactions without exponential approximations.

**Catalyst Design:** Modeling energy profiles for clean fertilizer production.

**Battery Chemistry:** Simulating solid-state electrolyte stability.

**Drug Discovery:** Calculating precise binding affinities for targeted therapeutics.

APPLICATION DOMAIN

# Combinatorial Optimization and Logistics

Solving high-dimensional NP-hard scheduling bottlenecks across global networks.

**Supply Chains:** Dynamic routing across constrained transit hubs.

**Port Terminals:** Real-time container sequencing to relieve ship queues.

**Energy Grids:** Instantaneous load balancing across renewable power sources.



APPLICATION DOMAIN

# Quantitative Finance and Risk Analytics

### Portfolio Hedging

Quadratic acceleration in value-at-risk calculations.

### Arbitrage Finding

Real-time cycle detection across high-frequency pairs.

### Fraud Detection

Quantum kernel methods for correlated anomaly recognition.

EMERGING FRONTIER

# Quantum Computing and Generative AI

### Parameterized Quantum Circuits

**Native Sampling:** Efficient generation of complex quantum probability distributions.

**Hybrid Workflows:** Classical optimizers tuning quantum gate parameters.

### Quantum Machine Learning

**Hilbert Spaces:** Projecting data into high-dimensional quantum states.

**Pattern Recognition:** Detecting correlations inaccessible to classical neural nets.

WELCOME TO DAY 1

# Come Curious. Leave Inspired.

Explore the foundations, build with Qiskit, and collaborate during Fall Fest 2026.

Qiskit KNUST Student Chapter

# Image Sources



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Source: calisphere.org

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